

Orbital Stewardship: A Framework for a Collective Liability Fund to Combat Space Debris

Abstract¹

The proliferation of orbital debris presents an escalating threat to the sustainability of space activities critical to global economy and international security. The current international legal framework was established before the scale of the debris problem was fully understood. Further, it does not account for modern debris tracking and recovery mechanisms. These are considerations that are necessary in constructing an effective liability regime for space debris, a regime that recognizes both the importance of prevention of debris formation as well as developments in active debris removal (ADR). The present regime's reliance on proving direct causation and fault for orbit incidents is unworkable in an environment characterized by stochastic collision events. This article proposes a paradigm shift away from this failing model of *post facto* blame attribution towards a system of *ex ante* risk quantification and collective financial responsibility. This framework outlines the structure for an Orbital Debris Compensation Fund (ODCF), an international, actuarial style risk pool designed to internalize the costs of creating orbital hazards. The model's foundation is a data driven risk assessment methodology that centres around modern space situational awareness capabilities.

Introduction

By integrating high fidelity data from sources such as the European Space Agency's (ESA) DISCOS database and the U.S. Space Force's Space Fence radar system, the model can assign a composite Orbital Risk Score (ORS) to every object in orbit. The ORS is a weighted metric calculated from key risk driving parameters, including an object's debris generating potential (mass, construction, stored energy), its orbital persistence (orbital lifetime), and its collision probability (determined by the traffic density of its orbital region). An operator's contribution to the ODCF would be directly proportional to their share of the total aggregated risk in orbit, as measured by the sum of their objects' ORS scores. This approach has been followed principally in other international compensation regimes in the maritime and nuclear sectors, creating tangible incentives for sustainable behaviour. Operators have complete control over their contribution to the model by implementing debris conscious design choices, complying with post mission disposal guidelines, and deorbiting space objects in a timely manner, as the model creates financial incentive for sustainable debris practices. The ODCF would serve two primary functions,

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providing no fault compensation for damages caused by untraceable debris and funding proactive Active Debris Removal (ADR) missions. By creating a market for ADR and prioritizing targets based on their ORS, the fund can significantly address the space debris problem through both mitigatory and active removal means.

The idea of pooling responsibility for orbital debris is not new, and this article builds upon it rather than inventing it. Commentators have for decades proposed collective solutions to the problem of untraceable debris—international compensation funds or “liability pools” to which launching entities would contribute, and market-share liability regimes adapted from terrestrial mass-tort law.² What these proposals have generally lacked is a rigorous, data-driven basis for apportioning contributions among operators in a way that reflects the risk each actually imposes. The contribution of this article is to supply that missing element: an actuarial metric, the Orbital Risk Score, that ties each operator’s share of the fund directly to the quantified hazard its objects create in the orbital commons.

This article provides a technical and legal blueprint for this framework, offering a viable path toward ensuring the long term stewardship and sustainability of the orbital commons. For this purpose, it is structured into five parts. The first part establishes the need for an alternative, actuarial approach to space debris liability by demonstrating how the current framework’s conceptual and legal limitations fail to effectively address the debris problem. The second part lays out the Space Situational Awareness (SSA) data on the basis of which the model is designed, by envisioning an integration between different existing data systems, as well as addressing the political motivations and incentives that make such international collaboration feasible. The third part defines the basis of the model, the Orbital Risk Score (ORS). The ORS is a single metric combining debris generation, orbital persistence, and collision probability, directly tying each operator’s risk to financial liability. The fourth part designs the financial and structural architecture of the Orbital Debris Compensation Fund (ODCF), detailing its contribution formula on the basis of the ORS, its internal governance, no fault claims process, as well as envisioning a phased implementation plan. Lastly, the fifth part explains how the ODCF fosters a market for Active

² Such collective approaches have a substantial pedigree. See Mark J. Sundahl, *Unidentified Orbital Debris: The Case for a Market-Share Liability Regime*, 24 *Hastings Int’l & Comp. L. Rev.* 125 (2000); Armel Kerrest, *Space Debris: Notion of “Space Object”* (proposing an international fund, financed in proportion to debris creation, to compensate damage caused by unidentified debris); Luke Punnakanta, *Space Torts: Applying Nuisance and Negligence to Orbital Debris*, 86 *S. Cal. L. Rev.* 163, 185–90 (2012) (canvassing the liability-pool and market-share proposals).

Debris Removal (ADR), using ORS based prioritization and funding mechanisms to create financial incentives for both public and private cleanup of high risk debris.

I. THE FAILURES OF THE TRADITIONAL LIABILITY MODEL

A. The Tragedy of the Commons in Low Earth Orbit

The near Earth space environment, particularly Low Earth Orbit (LEO), functions as a global commons, a shared resource vital for communications, navigation, Earth observation, and security. Like other common human resources, it is susceptible to the *tragedy of the commons*, where individual actors, operating in their own self-interest, degrade the resource for all.³ The creation of space debris is a classic negative externality: the cost of launching a satellite and potentially creating debris is not fully borne by the launching entity but is displaced onto all other orbital operators, a diffuse cost that no existing legal mechanism requires the debris-creator to internalise.⁴⁵

This long-standing issue has been dramatically amplified by the modern space era, characterized by the deployment of large commercial satellite mega constellations.⁶ These constellations, planned by entities such as SpaceX's Starlink, OneWeb, and Amazon's Project Kuiper, involve the launch of tens of thousands of satellites into already congested LEO bands.⁷ For instance, Starlink alone may eventually feature up to 42,000 satellites.⁸ This rapid increase in the number of satellites, coupled with the increased participation of largely unregulated private entities in outer space activities,⁹ is transforming orbital congestion from a theoretical future risk into a persistent, current operational hazard.¹⁰ The sheer density of these new constellations exponentially increases

³ Daniel Lambach & Luca Wesel, *Tackling the Space Debris Problem: A Global Commons Perspective*, 8th Eur. Conf. on Space Debris 165 (2021).

⁴ On orbital debris as an uninternalized negative externality, see Alexander William Salter, *Space Debris: A Law and Economics Analysis of the Orbital Commons*, 19 Stan. Tech. L. Rev. 221 (2016); Akhil Rao & Giacomo Rondina, *The Economics of Orbit Use: Open Access, External Costs, and Runaway Debris*. The cost is not "socialised" in any redistributive sense; it is displaced onto third-party operators because no rule currently requires the debris-creator to bear it.

⁵ Ibid

⁶ Sheer Abbas, *Challenges to Space Activities in the Context of Mega Satellite Constellations: A Focus on Environmental Impacts*, 42 J. Astronomy & Space Sci. 1 (2025).

⁷ Ereza Pultarova, *SpaceX Starlink Satellites Made 50,000 Collision Avoidance Maneuvers in the Past 6 Months. What Does That Mean for Space Safety?*, Space.com (23 July 2024), <https://www.space.com/spacex-starlink-50000-collision-avoidance-maneuvers-space-safety>. Kuiper Constellation 153 Satellites, SatelliteMap.space (15 Oct. 2025), <https://satellitemap.space/constellation/kuiper>.

⁸ Pultarova, *supra* n. 4.

⁹ Yuree Nam, Note, *One-Way Ticket to Mars: The Privatization of the Space Industry and its Environmental Impact on Earth and Beyond*, 19 Nw. J.L. & Soc. Pol'y 150 (2023)

¹⁰ Jonathan C. McDowell, *General Catalog of Artificial Space Objects, Release 1.8.0*, planet4589.org (Aug. 2020),

the frequency of conjunction alerts, necessitating a far higher rate of collision avoidance manoeuvres (CAMs).¹¹ [add to this now one million satellites proposed by SpaceX for data constellations!]

Satellite operators now bear tangible economic burdens from the constant need to perform CAMS, a direct cost imposed by the polluted environment.¹² These costs are multi-faceted. First, each manoeuvre consumes propellant, which directly shortens the operational lifetime of a satellite and reduces its long term revenue generating potential.¹³ Second, the process requires significant manpower for risk analysis, manoeuvre planning, and inter operator coordination, which can be a slow and manual process.¹⁴ Reports indicate that a single emergency manoeuvre in LEO can cost an estimated EUR 25,000, and a modest 300 satellite constellation could face approximately €14 million per year in associated staff and analysis costs.¹⁵ The scale of this issue is highlighted by data from SpaceX, which reported its Starlink satellites performed nearly 50,000 CAMs in a six month period from late 2023 to mid 2024, a number that had been doubling every six months previously.¹⁶ These manoeuvres also cause operational downtime, interrupting or delaying the satellite's primary mission, such as scientific observation or data transmission. The economic impact of this externality is therefore no longer abstract; it is a direct, recurring, and escalating operational expenditure for all actors in LEO.¹⁷ Inaction actively courts the Kessler Syndrome, a cascading chain reaction of collisions that could render key orbital bands unusable for generations.¹⁸

<https://planet4589.org/space/gcat>.

¹¹ European Space Agency, *Automating Collision Avoidance* (22 Oct. 2019), https://www.esa.int/Space_Safety/Space_Debris/Automating_collision_avoidance.

¹² Juan Luis Gonzalo et al., *Analytical Framework for Space Debris Collision Avoidance Maneuver Design*, 44 J. Guidance, Control & Dynamics 469 (2021).

¹³ *ibid.*

¹⁴ Thomas J. Colvin et al., NASA Office of Tech., Policy & Strategy, *Cost and Benefit Analysis of Orbital Debris Remediation* (Mar. 2023).

¹⁵ *ibid.*

¹⁶ *SpaceX's Semi-Annual Update on Starlink Network Health, Failure Rate, Collision Risk: Why Don't All Constellations Provide This Data?*, Kratos Space (13 Oct. 2025), <https://www.kratospace.com/constellations/articles/spacex-semi-annual-update>. *Collision Avoidance Maneuvers by Starlink Satellites Increasing Exponentially*, AIAA (7 July 2023), <https://aiaa.org/2023/07/07/collision-avoidance-maneuvers>.

¹⁷ Zeno Pavanello et al., *CAMmary: A Review of Spacecraft Collision Avoidance Manoeuvre Design Methods*, 236 Acta Astronautica (2025). Akhil Rao & Giacomo Rondina, *The Economics of Orbit Use: Open Access, External Costs, and Runaway Debris Growth*, 12 J. Ass'n Env'tl. & Res. Economists 353 (2025).

¹⁸ Elad Denenberg & Pini Gurfil, *Debris Avoidance Maneuvers for Spacecraft in a Cluster*, 40 J. Guidance, Control & Dynamics 1428 (2017). *Simulating The Kessler Syndrome: How Satellite Collisions Could Spiral Out of Control*, Science Buddies (15 Oct. 2025), https://www.sciencebuddies.org/science-fair-projects/project-ideas/SpaceEx_p058/space-exploration/kessler-syndrome.

B. The Kessler Syndrome

The scientific understanding of debris proliferation has evolved from simple linear projections to complex, non-linear models that describe a self-sustaining cascade. Donald J. Kessler first posited that once the spatial density of objects in a given orbital region reaches a critical density, the rate of new debris generation from collisions will exceed the rate at which objects are naturally removed by forces like atmospheric drag.¹⁹ Beyond this point, the debris population becomes self-propagating, destined to increase even if all launch activities were to cease immediately. Scientific analysis indicates that key regions of LEO are already at or beyond this critical density, meaning the collisional cascade is no longer a future possibility but a pressing reality.²⁰

A simplified mathematical illustration of this dynamic, developed by Farinella and Cordelli, models the coupled evolution of two populations—large objects such as satellites and the small, lethal fragments produced when they collide.²¹ The detail of the equations need not detain us. The essential, well-established result is qualitative: once the rate at which collisions generate new fragments outpaces the rate at which fragments are added by launches or removed by atmospheric drag, the fragment population grows quasi-exponentially and, in time, drives the operational satellite population toward collapse.²² The lesson for the law is therefore not the precise figures but the character of the risk: orbital hazard is systemic, stochastic and non-linear, and it cannot be reduced to discrete, pairwise cause and effect.

C. The Legal Impasse: The Limits of the Liability Convention

The international liability regime for space activities rests on two instruments, and its structure is the analytical hinge of the entire debris problem. Article VII of the Outer Space Treaty lays down the basic rule that a launching State “is internationally liable for damage” caused by its space object to another State Party or its nationals “on the Earth, in air space or in outer space.”²³ The 1972 Liability Convention elaborates that rule into a deliberately bifurcated standard. For damage occurring on the surface of the Earth or to aircraft, Article II imposes absolute liability: “A

¹⁹ Donald J. Kessler & Burton G. Cour-Palais, *Collision Frequency of Artificial Satellites: The Creation of a Debris Belt*, 83 J. Geophysical Res.: Space Physics 2637 (1978).

²⁰ OECD, *Earth's Orbits at Risk* (14 Sept. 2022), https://www.oecd.org/en/publications/earth-s-orbits-at-risk_16543990-en.html.

²¹ A. Cordelli et al., *The Influence of the Fragmentation Threshold on the Long-Term Evolution of the Orbital Debris Environment*, 46 Planetary & Space Sci. 691 (1998).

²² *ibid.*

²³ Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space art. VII, Jan. 27, 1967, 18 U.S.T. 2410, 610 U.N.T.S. 205 [Outer Space Treaty].

launching State shall be absolutely liable to pay compensation for damage caused by its space object on the surface of the Earth or to aircraft in flight.”²⁴ For damage occurring in orbit, however, Article III imposes only fault liability, providing that where damage is “caused elsewhere than on the surface of the Earth” to one space object by another, the launching State “shall be liable only if the damage is due to its fault or the fault of persons for whom it is responsible.”²⁵ This bifurcation is decisive. The catastrophic and recurring harm at the centre of the debris crisis—collision between objects in orbit—is precisely the category consigned to the most demanding standard, fault; the claimant-friendly absolute standard is confined to the comparatively rare event of debris surviving re-entry to injure persons or property on Earth. The remainder of this Part shows why the fault standard is, in the orbital environment, practically impossible to satisfy.

That a victim of an in-orbit collision must establish fault carries consequences the drafters cannot have foreseen. Fault liability, in every major legal tradition, imports a causation requirement: the claimant must connect a specific wrongful act to a specific harm through a sufficiently direct causal link. Common-law systems express this through the doctrine of proximate cause; civil-law systems through cognate concepts such as adequate causation. The precise formulation varies across legal systems, but the demand for a traceable, unbroken chain between wrongdoer and injury is common to them all²⁶—and it is exactly that chain which the debris environment severs. Two features of the regime make the fault standard practically unmeetable. First, fault presupposes the breach of an established legal duty or standard of conduct.²⁷ Yet the principal benchmarks for debris mitigation—the UNCOPUOS Space Debris Mitigation Guidelines and the IADC Guidelines from which they derive—are, by their own terms, voluntary and non-binding.²⁸ Because they do not crystallise into a firm, obligatory norm of conduct, a State’s failure to observe them does not, without more, establish fault: there is no binding rule against which the State’s behaviour can be measured as legally wrongful, and a State may decline to passivate a rocket body

²⁴ Convention on International Liability for Damage Caused by Space Objects art. II, Mar. 29, 1972, 24 U.S.T. 2389, 961 U.N.T.S. 187 [Liability Convention].

²⁵ Liability Convention, *supra*, art. III.

²⁶ On the causation requirement in fault liability, see Restatement (Second) of Torts §§ 430–453 (Am. L. Inst. 1965) (duty, breach, and proximate cause); the civil-law analogue is the doctrine of adequate causation (*adäquate Kausalität*). The formulations differ, but each requires a sufficiently direct causal link between the wrongful act and the harm.

²⁷ Convention on International Liability for Damage Caused by Space Objects art. III, Mar. 29, 1972, 24 U.S.T. 2389, 961 U.N.T.S. 187.

²⁸ UNCOPUOS, *Compendium: Space Debris Mitigation Standards Adopted by States and International Organisations*, U.N. Doc. A/AC.105/C.2/2021/CRP.19 (June 2021).

or to deorbit a defunct satellite without breaching any hard obligation.²⁹ To be sure, a growing body of scholarship argues that mitigation is hardening into a customary obligation, or that it is already entailed by the due-regard duty of Article IX of the Outer Space Treaty; but that position remains contested, and its very contestation is the point, for the content of the supposed duty is unsettled.³⁰

Secondly, even where a duty could be established, the regime's insistence on identifying a particular launching State collapses in the face of what may be called the debris-from-debris problem.³¹ Consider a fragment from Satellite A, launched decades ago, that strikes Satellite B and produces a cloud of new fragments; one of these, Fragment C, later strikes Satellite D, generating a further cloud; and a piece from that second cloud ultimately disables a victim's operational satellite. Under an orthodox proximate-cause analysis, the intervening collisions involving B and D would be treated as superseding intervening causes, severing the legal chain back to the launcher of Satellite A.³² The Kessler cascade is, in effect, a continuous sequence of such intervening events, making it both scientifically and legally impossible to trace a given fragment, through successive collisions, to a single responsible State.³³ This is not to overstate the difficulty. Not all debris-on-debris damage is untraceable: large fragments are catalogued and tracked, and modern space situational awareness has permitted confident attribution of discrete events—most notably the 2009 collision between the defunct Cosmos 2251 and the operational Iridium 33.³⁴ The intractable cases are the two that the debris crisis is steadily making the norm: damage caused by small, untracked fragments below the catalogue threshold, and damage at the end of a multi-generational cascade in which attribution dissolves across successive intervening collisions.

This evidentiary difficulty is compounded by a foundational ambiguity in the treaties themselves. The Outer Space Treaty and the Liability Convention regulate “space objects” and their

²⁹ *ibid.*

³⁰ For the contested view that debris mitigation is hardening into a binding obligation—whether as custom or as a corollary of the Article IX due-regard duty—see Lawrence Li, *Space Debris Mitigation as an Obligation*, 17 *Int'l Cmty. L. Rev.* 297 (2015).

³¹ *Liability Convention*, *supra* n. 25, at art. III.

³² On superseding intervening causes severing the chain of legal causation, see Restatement (Second) of Torts §§ 440–442 (Am. L. Inst. 1965).

³³ Kessler & Cour-Palais, *supra* n. 21.

³⁴ The 2009 collision was promptly and confidently attributed because both objects were catalogued and tracked. See Ram Jakhu, *Iridium-Cosmos Collision and its Implications for Space Operations*, 52 *Proc. Colloquium L. Outer Space* 254 (2009).

“component parts” but nowhere define those terms by reference to debris.³⁵ The Convention provides only that the term “space object” “includes component parts of a space object as well as its launch vehicle and parts thereof.”³⁶ Applying the interpretive rule of Article 31 of the Vienna Convention on the Law of Treaties—ordinary meaning, in context, in the light of the treaty’s object and purpose³⁷—the better reading is that an intact defunct satellite, and probably its identifiable large fragments, remain space objects or component parts within the Convention, since the object and purpose of the regime is to secure compensation for space-caused damage. The difficulty lies at the margin, and the margin is where most of the lethal population sits. It is genuinely uncertain whether a paint fleck, a frozen coolant droplet, or a millimetre-scale fragment liberated in a collision continues to qualify as a component part of its progenitor spacecraft. Kerrest, among others, has questioned whether a piece that has detached from its source and serves no function can still be characterised as a component part at all, and a further strand of the literature argues that debris should be regulated as a category distinct from space objects altogether.³⁸ If that narrower reading prevails, a vast and lethal portion of the debris population falls outside the Convention entirely.³⁹ The consequence is a perverse, tiered structure of legal risk. A large, identifiable defunct satellite is plainly a space object over which the State of registry retains jurisdiction and control under Article VIII of the Outer Space Treaty,⁴⁰ and for which a launching State remains liable. But the legal status of the thousands of untraceable fragments that such an object can generate upon break-up is contestable—so that the most probable cause of satellite failure, impact by small untrackable debris, is also the least legally actionable, because a launching State can plausibly argue that such fragments are not component parts and so fall outside the

³⁵ Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies, Jan. 27, 1967, 18 U.S.T. 2410, 610 U.N.T.S. 205.

³⁶ Liability Convention, *supra*, art. I(d) (“The term ‘space object’ includes component parts of a space object as well as its launch vehicle and parts thereof.”). An identical definition appears in art. I(b) of the Registration Convention.

³⁷ Vienna Convention on the Law of Treaties art. 31(1), May 23, 1969, 1155 U.N.T.S. 331 (“A treaty shall be interpreted in good faith in accordance with the ordinary meaning to be given to the terms of the treaty in their context and in the light of its object and purpose.”).

³⁸ See Armel Kerrest, *Space Debris: Notion of “Space Object”* (questioning whether a fragment that has detached from its source and serves no function remains a “component part”); Frans G. von der Dunk, *Space Debris and the Law* (noting the unresolved question whether debris falls within “space object” for purposes of the Liability Convention). A further strand argues that debris should be regulated as a category distinct from space objects altogether.

³⁹ Luke Punnakanta, Note, *Space Torts: Applying Nuisance and Negligence to Orbital Debris*, 86 S. Cal. L. Rev. 163 (2012). Peter Stubbe, *State Accountability for Space Debris: A Legal Study of Responsibility for Polluting the Space Environment and Liability for Damage Caused by Space Debris* (Brill Nijhoff 2017).

⁴⁰ Outer Space Treaty, *supra*, art. VIII (the State of registry “shall retain jurisdiction and control over such object”). The correct treaty concept is jurisdiction and control, not “jurisdiction and liability.”

Convention's reach.⁴¹

D. The Policy Imperative for a New, Risk Based Model

The disconnect between legal doctrine and physical reality compels a change of approach. The law seeks a linear chain of causation; the science presents a probabilistic web of interaction. The paradigm the regime was built to handle is illustrated by the case that, in practice, prompted much of its drafting—the 1978 re-entry of the Soviet nuclear-powered satellite Cosmos 954 over the Northwest Territories of Canada. That episode had two features the present crisis lacks. First, attribution was straightforward: the object was large, tracked, and its Soviet provenance undisputed, so there was no causal mystery to solve.⁴² Second, because the damage occurred on the surface of the Earth, it fell squarely within the absolute-liability standard of Article II, and Canada was therefore relieved of any need to prove fault; the claim was ultimately resolved by a negotiated payment of three million Canadian dollars.⁴³ The regime functioned precisely because the case combined clear attribution with surface damage—the two conditions that in-orbit debris collisions systematically negate.⁴⁴ Where harm is instead caused in orbit by an untraceable fragment, the claimant faces the worst of both worlds: the demanding fault standard of Article III and an impossible burden of proof. The doctrinal response must therefore mirror the structure of the problem, shifting away from the post hoc attribution of blame for discrete incidents and toward the ex ante quantification and allocation of responsibility for contributions to systemic risk. That move has a respectable pedigree in terrestrial law, where doctrines such as market-share liability were developed to apportion responsibility for harms caused by diffuse, multi-source pollution among the actors who created the risk.⁴⁵ A collective, no-fault compensation mechanism, funded in proportion to the risk each operator imposes, is the logical extension of that approach to the orbital commons.⁴⁶

⁴¹ Alexander William Salter, *Space Debris: A Law and Economics Analysis of the Orbital Commons*, 19 Stan. Tech. L. Rev. 221 (2016).

⁴² Kessler & Cour-Palais, *supra* n. 21.

⁴³ Protocol Between the Government of Canada and the Government of the Union of Soviet Socialist Republics Concerning the Settlement of Canada's Claim for Damages Caused by the Satellite "Cosmos 954," Apr. 2, 1981, 20 I.L.M. 689 (settlement of three million Canadian dollars). Because the damage occurred on the surface of the Earth, Canada's claim sounded in the absolute-liability standard of art. II and did not require proof of fault.

⁴⁴ Ram Jakhu, *Iridium-Cosmos Collision and its Implications for Space Operations*, 52 Proc. Colloquium L. Outer Space 254 (2009).

⁴⁵ Anthony M. Marino, *Market Share Liability and Economic Efficiency*, 57 S. Econ. J. 667 (1991).

⁴⁶ Mark J. Sundahl, Note, *Unidentified Orbital Debris: The Case for a Market-Share Liability Regime*, 24 Hastings Int'l & Comp. L. Rev. 125 (2000).

II. Data Set for Risk-Based Model

The technical prerequisites for a data driven, actuarial style liability model are now being met by modern space situational awareness (SSA) systems. A robust risk model requires a system of systems approach, fusing data from multiple, complementary sources to create a comprehensive and dynamic catalogue of the orbital environment.

A. ESA's DISCOS Database

The European Space Agency's (ESA) Database and Information System Characterizing Objects in Space (DISCOS) serves as a foundational catalogue of the known, trackable space object population, providing a single source reference for all unclassified objects.⁴⁷ Its architecture and data are ideally suited to form the backbone of the proposed risk model. DISCOS uses two critical, persistent identifiers for each catalogued object; the International Designator, or *COSPAR ID*, which links an object to its specific launch event (e.g., 1998 067A for the International Space Station), and the *NORAD ID* (also known as the Satellite Catalog Number or SATNO), a unique sequential number assigned by the U.S. Space Command to each object tracked. These identifiers are essential as they provide a universal key for correlating information about a single object across multiple databases and sensor networks. The database contains detailed physical characteristics crucial for risk assessment, including an object's mass, shape, and dimensions (length, height, depth), as well as computed values like its average cross sectional area (xSectAvg).⁴⁸ Furthermore, its detailed typology, which classifies objects as *Payload*, *Rocket Body*, or various forms of debris such as *Payload Fragmentation Debris*, provides vital information for assessing an object's inherent risk of spontaneous breakup.⁴⁹ The most critical technical enabler for the proposed framework is that DISCOS data is accessible programmatically via a Representational State Transfer (REST) Application Programming Interface (API).⁵⁰ This machine to machine interface allows for the automated, scalable, and continuous ingestion of object attribute data into an external risk modelling engine, transforming the problem from one of impossible manual data collection to one of programmatic data integration and making a global, dynamic risk assessment system technically feasible.⁵¹

⁴⁷ *DISCOSweb*, ESA (15 Oct. 2025), <https://discosweb.esoc.esa.int>.

⁴⁸ *DISCOSweb Operations API How-To*, ESA (15 Oct. 2025), <https://discosweb-api.sdo.esoc.esa.int>.

⁴⁹ H. Klinkrad, *DISCOS—ESA's Database and Information System Characterising Objects in Space*, 11 *Advances Space Res.* 43 (1991).

⁵⁰ *ibid.*

⁵¹ Quirin Funke et al., *An Access Point to ESA's Space Debris Data: The Space Debris Office Web-Based Tools* (6th Int'l Conf. on Astrodynamics Tools & Techniques, 2016).

B. The U.S. Space Fence: High Fidelity Tracking of the Debris Environment

While DISCOS provides detailed information on known objects, the U.S. Space Force's Space Fence system provides the high frequency, high resolution tracking necessary to monitor the dynamic orbital environment, including previously untrackable objects.⁵² As a second generation SSA radar system operating in the S band, Space Fence's key capability is uncued detection, meaning it can discover and track objects without prior knowledge of their orbit.⁵³ This allows it to catalogue a much larger population of smaller, lethal debris, down to objects roughly 10 cm across in LEO, that pose a significant threat but are too small for legacy systems to track.⁵⁴ The system's capabilities represent a quantum leap in SSA data fidelity, making over 1.5 million observations per day and tracking an estimated 200,000 objects, which constitutes a tenfold increase over previous systems.⁴⁴ Its equatorial location provides surveillance coverage of objects in LEO, Medium Earth Orbit (MEO), and Geostationary Orbit (GEO).⁵⁵ Critically, the system is built on a net centric architecture designed to share data with authorized users, such as the Joint Space Operations Center (JSpOC), via web services and standard internet protocols.⁵⁶ This programmatic data sharing capability is the essential link that allows its high precision positional data to be fused with the object attributes from DISCOS, providing the real time inputs needed for an accurate and dynamic risk model.

C. Data Fusion and API Integration: Creating a Unified Data Layer for Risk Assessment

The true power of modern SSA lies not in a single system, but in the fusion of an historical, attribute-rich catalogue (DISCOS) with a real time, high-resolution sensor network (Space Fence). A unified data layer, constructed via Application Programming Interface (API) integration, is essential for a credible risk model. This approach is necessary to overcome the significant limitations of legacy data models, which are predominantly based on Two Line Element sets (TLEs).⁵⁷ TLEs rely on simplified general perturbation models that are known to produce

⁵² *Swinging for the Space Fence*, U.S. Space Force (7 Apr. 2020), <https://www.spaceforce.mil/News/Article-Display/Article/2142648/swinging-for-the-space-fence>.

⁵³ G.P. Fonder et al., *AN/FSY-3 Space Fence System—Sensor Site One/Operations Center Integration Status and Sensor Site Two Planned Capability* (Advanced Maui Optical & Space Surveillance Tech. Conf., 2017).

⁵⁴ *ibid.*

⁵⁵ *Debris Detection Developments—The Space Fence*, MilsatMagazine, <http://www.milsatmagazine.com/story.php?number=1554778712>.

⁵⁶ *Swinging for the Space Fence*, *supra* n. 43; *Space Fence: Surveillance System*, Lockheed Martin, <https://www.lockheedmartin.com/en-us/products/space-fence.html>.

⁵⁷ Pavel Kovář et al., *Study of the Two-Line Element Accuracy by 1U CubeSat with a GPS Receiver*, 22 Sensors (Basel) 2902 (2022).

significant inaccuracies, with positional errors that can be on the order of kilometres, particularly for small objects in LEO that are subject to substantial and variable atmospheric drag.⁵⁸ Such low fidelity data is fundamentally unsuitable for the precise risk calculations required by an actuarial framework. In contrast, modern systems like Space Fence provide high fidelity state vectors, precise position and velocity data, along with associated covariance matrices that quantify the uncertainty in those measurements.⁵⁹

The proposed fusion process leverages these complementary strengths. For any given catalogued object, identified by its universal NORAD ID, the risk model's engine would execute two programmatic calls. First, it would query the Space Fence network via its web services to retrieve the object's precise, real time state vector, answering the question of *where* it is with high accuracy. Second, it would query the DISCOS database via its REST API to retrieve the object's detailed physical profile, answering the question of *what* it is (mass, size, owner, type). For the vast population of smaller debris detected by Space Fence but absent from the DISCOS catalogue, the high volume of observational data is used to build and continuously update a dynamic spatial density map of the orbital environment. This map becomes a direct, measurement driven input for calculating the collision probability for any satellite transiting a given orbital region, transforming this aspect of risk assessment from a static estimate to a near real time calculation.

D. Plausibility of Interagency Collaboration

The technical feasibility of integrating data from systems operated by different international bodies, such as ESA and the U.S. Space Force, is underpinned by a strong history of collaboration and a compelling common interest. There is a robust precedent for technical cooperation between these agencies, most notably through the Inter Agency Space Debris Coordination Committee (IADC). The IADC, which includes ESA, NASA, and other major global space agencies, has served for decades as the primary international forum for developing consensus on technical standards for debris mitigation, demonstrating a long and successful history of collaboration on this exact issue.⁶⁰ Furthermore, the shared and indiscriminate threat that orbital debris poses to all space actors creates a powerful incentive for collective action. Although outer space is not the “common heritage of mankind”—a designation that appears only in the sparsely ratified Moon

⁵⁸ David Vallado & Paul Cefola, *Two-Line Element Sets—Practice and Use* (7th Int'l Astronautical Cong., 2012).

⁵⁹ Daniel L. Pechkis et al., *Statistical Approach to the Operational Testing of Space Fence* (Advanced Maui Optical & Space Surveillance Tech. Conf., 2014).

⁶⁰ IADC, *IADC Report on the Status of the Space Debris Environment*, IADC-23-01 Issue 3 (2025).

Agreement and nowhere in the Outer Space Treaty⁶¹—Article IX of the Outer Space Treaty independently supplies a legal foundation for cooperative debris governance. It requires every State to conduct its activities “with due regard to the corresponding interests of all other States Parties” and to avoid the “harmful contamination” of space, and, properly construed, that obligation does not merely permit cooperation but presses States toward active measures to limit the debris they generate, since the unchecked creation of collision hazards cannot easily be squared with due regard for every other user of the orbit.⁶² A runaway collisional cascade in the LEO would be devastating to the operational and economic interests of ESA, NASA, and commercial operators alike, making the creation of a unified risk assessment model a matter of shared survival rather than national competition. The existing operational practice of the U.S. Space Surveillance Network sharing conjunction data with ESA for collision avoidance purposes already provides a direct precedent for the type of data exchange that would be required.⁶³ While political and security concerns over sharing raw sensor data from national security assets are valid, the proposed ODCF framework can be designed to mitigate these issues. The fund’s administrator could operate as a trusted, neutral third party, receiving sensitive data under strict confidentiality agreements. Alternatively, national SSA providers could use a standardized, transparent algorithm to calculate the ORS for their registered objects internally and then report only the final, non-sensitive risk scores to the fund. This approach respects national security boundaries while still enabling the necessary data aggregation for the fund to function, making broad international collaboration politically plausible.

III. The Orbital Risk Model: An Actuarial Approach

Core Principles From Individual Causation to Probabilistic Risk Contribution

The proposed model shifts the legal and financial paradigm from proving causation for individual collisions to quantifying each object’s probabilistic contribution to systemic risk. This methodology is directly analogous to actuarial science, which uses historical data and predictive

⁶¹ Agreement Governing the Activities of States on the Moon and Other Celestial Bodies art. 11, Dec. 18, 1979, 1363 U.N.T.S. 3. The Moon Agreement is the only United Nations space treaty to employ the phrase “common heritage of mankind,” and it has been ratified by relatively few States, none of them major spacefaring powers. The term does not appear in the Outer Space Treaty.

⁶² Outer Space Treaty, *supra*, art. IX (“due regard to the corresponding interests of all other States Parties”; duty to avoid “harmful contamination”). See also Lawrence Li, Space Debris Mitigation as an Obligation, 17 Int’l Cmty. L. Rev. 297 (2015) (locating a mitigation duty within the due-regard obligation).

⁶³ Funke et al., *supra* n. 42.

models to assess risk and determine insurance premiums.⁶⁴ The objective is not to assign blame for a past event but to assign a price to the future risk an object imposes on the orbital commons. This is accomplished by calculating a composite Orbital Risk Score (ORS) for every object, translating complex astrodynamics and engineering characteristics into a single, quantifiable metric.

The ORS is derived from three primary, weighted risk factors, each composed of several sub factors.

i. Debris Generation Potential (DGP)

This factor quantifies an object's potential to create new debris upon collision or breakup. It is a composite score based on the following physical and operational characteristics, with each sub factor being inferred from the fused SSA data layer and interpreted through established scientific models.

The **Mass (M)** of an object is the primary driver of the quantity of fragments generated in a catastrophic collision, which is defined as an impact with an energy to mass ratio of 40 J/g or greater (if the energy to mass ratio is less than 40 J/g, the collision is not 'catastrophic', as the object is likely to suffer cosmetic non-severe damage, as opposed to fragmentation).⁶⁵ Scientific analysis, particularly NASA's Standard Breakup Model, demonstrates that the number of fragments (N) larger than a characteristic length (L_c) scales with the target's mass according to a power law, commonly expressed as $N(L_c) = 0. (M)^{0.75} L_c^{-1.71}$.⁵⁹ Simply put, the more the mass of an object, the more fragments it generates upon catastrophic collision. For the purposes of the ORS calculation, this mass value is sourced directly from the mass field for the corresponding object in the DISCOS database.⁶⁶

The **Collision Cross Section (A_c)** represents an object's average physical area perpendicular to the direction of a potential impact. A larger cross sectional area directly and linearly increases the probability of an object being struck. This parameter is sourced from the *xSectAvg* field in the DISCOS database, which provides a computed average cross section based on the object's known

⁶⁴ 'Actuarial Models, EBSCO Research, <https://www.ebsco.com>.

⁶⁵ Gene Stansbery et al., *A Comparison of Catastrophic On-Orbit Collisions* (Advanced Maui Optical & Space Surveillance Tech. Conf., 2008).

⁶⁶ *DISCOSweb Operations API How-To*, *supra* n. 39.

dimensions and shape.⁶⁷

The **Fragmentation Potential (FP)** is a categorical score that quantifies the likelihood of an object breaking up due to internal causes, such as a spontaneous explosion from stored energy. Objects with significant residual propellants, pressurized tanks, or charged batteries that have not been passivated at the end of their mission pose a much higher risk of spontaneous fragmentation.⁶⁸ This risk is inferred primarily from the object's classification in the DISCOS database; for example, an object with an *objectClass* of 'Rocket Body' is assigned a high FP score, whereas a modern satellite designed with end of life passivation protocols receives a low score.⁶⁹

The **Material Density (ρ_m)** of an object's constituent materials is a critical variable, as higher density fragments are more lethal upon impact, carrying more kinetic energy for a given size. To account for this, the ORS model assigns a risk multiplier based on the material composition categories defined in NASA's Orbital Debris Engineering Model (ORDEM).⁷⁰ ORDEM classifies debris populations into distinct categories such as low density (e.g., plastics, 1.4 g/cm³), medium density (e.g., aluminium, 2.8 g/cm³), and high density (e.g., steel, 8.0 g/cm³).⁷¹ An object's material composition can be inferred from its type and design, often using information available from tools like NASA's Debris Assessment Software (DAS), which mission planners use to verify compliance with mitigation standards.⁷²

ii. Orbital Persistence Index (OPI)

This factor measures the temporal aspect of risk, how long an object and any debris it might generate will remain in orbit, polluting the commons. A longer orbital lifetime equates to a higher risk contribution over time. The OPI is a direct function of the object's orbital lifetime (τ), which is calculated using well established orbital decay models, such as those developed by Desmond King Hele.⁷³ In the LEO, the primary force driving orbital decay is atmospheric drag. The models calculate orbital decay using two key variables; the object's altitude (h) and its area to mass ratio

⁶⁷ Klinkrad, *supra* n. 40.

⁶⁸ NASA APPEL, *Report on Orbital Debris Management & Risk Mitigation* (2023).

⁶⁹ Klinkrad, *supra* n. 40.

⁷⁰ Mark Matney et al., *ORDEM 4.0: NASA's Orbital Debris Engineering Model—A Status* (9th Eur. Conf. on Space Debris, 2025).

⁷¹ Mark Matney, *An Overview of NASA's Orbital Debris Environment Model* (NASA Orbital Debris Program Office 2010).

⁷² J. Opiela & N. Johnson, *Improvements to NASA's Debris Assessment Software* (58th Int'l Astronautical Cong., 2007).

⁷³ Nguyen X. Vinh et al., *Hypersonic and Planetary Entry Flight Mechanics* (Univ. Mich. Press 1980).

(A/m), a term often expressed as the inverse of the ballistic coefficient.⁷⁴ A ballistic coefficient is a measure combining an object's mass, size, and shape into a single number that quantifies the rate at which a space object slows down due to atmospheric drag. A high ballistic coefficient means the object is heavy and compact, it slows down slower and stays in orbit longer. Whereas a low ballistic coefficient means the object is light or has a large surface area, so it slows down faster and deorbits sooner. Therefore, the inverse of the ballistic coefficient is directly proportional to how quickly an object reenters the atmosphere.⁷⁵

The magnitude of atmospheric drag itself is determined by the atmospheric density at a given altitude, which is not constant. It is modelled using sophisticated atmospheric models, such as the Jacchia family of models, which account for variations in atmospheric density driven by solar activity.⁷⁶ Solar activity is measured by indices like the F10.7 solar radio flux, which is continuously monitored by agencies such as NOAA.⁷⁷ The data required for this calculation are drawn directly from the fused SSA data layer; the object's mass and average cross sectional area (used to calculate A/m) are retrieved from the DISCOS database, while its precise, real time orbital parameters, especially altitude, are provided by high fidelity tracking systems like the Space Fence.⁷⁸ Objects in higher LEO altitudes (e.g., above 800 km) or those with low area to mass ratios (i.e., dense and compact objects) have high ballistic coefficients, and therefore can have orbital lifetimes spanning centuries or even millennia, posing a persistent, long term threat and thus receiving a correspondingly high OPI score.⁷⁹

iii. Collision Probability Factor (CPF)

This factor accounts for the traffic density of an object's operational environment. An object in a crowded orbit poses a greater risk than an identical object in a sparsely populated region. The CPF is derived from the spatial density (ρ_s) and associated flux of debris for the specific orbital shell, defined by altitude and inclination, that an object occupies. This crucial environmental data is the direct output of sophisticated, validated engineering models like NASA's Orbital Debris Engineering Model (ORDEM) and ESA's Meteoroid and Space Debris Terrestrial Environment

⁷⁴ Sukdev Mahapatra, *Modeling Orbital Decay of Low-Earth Orbit Satellites due to Atmospheric Drag: A Simplified Analytical Approach*, Aug. 2025, astro-ph/2508.19549.

⁷⁵ *ibid.*

⁷⁶ L.L. Perini, Johns Hopkins Univ. Applied Physics Lab., *Orbital Lifetime Estimates*, ANSP-M-8 (1974).

⁷⁷ Bureau of Meteorology, *Satellite Orbital Decay Calculations* (Australian Gov't), <http://www.bom.gov.au>.

⁷⁸ Matney et al., *supra* n. 60.

⁷⁹ *ibid.*

Reference (MASTER) model.⁸⁰

These are not static, theoretical constructs; they are dynamic models that are continuously updated and validated with measurement data from the global Space Surveillance Network. This includes the high resolution data on the small but lethal debris population (1-10 cm) provided by the Space Fence, which dramatically improves the accuracy of the density maps in LEO.⁸¹ The ORS calculation process involves inputting an object's precise orbital state vector (sourced from a high fidelity tracker like Space Fence) into the ORDEM or MASTER model. The model then returns the expected flux of debris that the object will encounter in its specific orbital shell. This flux value is then normalized to assign the CPF. Consequently, objects operating in historically crowded and high value orbits, such as the sun synchronous orbits between 800-900 km, will have the highest debris density values and will therefore be assigned the highest CPF scores, reflecting their elevated risk profile.

iv. The Weighting Methodology, and the Orbital Risk Score (ORS) Formula

To create a single, actionable metric, these factors must be combined using a weighted scoring model, a common technique in multi variable risk assessment.⁸² The proposed Orbital Risk Score (ORS) would be calculated using a formula such as,

$$ORS = (w_{DGP} \cdot DGP_{norm}) + (w_{OPI} \cdot OPI_{norm}) + (w_{CPF} \cdot CPF_{norm})$$

where the individual factors (*DGP*, *OPI*, *CPF*) are normalized to a common scale, and the weights (w_{DGP} , w_{OPI} , w_{CPF}) sum to 1. In the implemented model, this weighted sum is further scaled by two gating factors, so that $ORS = R_{(hp)} \cdot A \cdot (w_{DGP} \cdot DGP + w_{OPI} \cdot OPI + w_{CPF} \cdot CPF)$, where *R* is a collision-relevance factor equal to 1 for objects whose perigee lies within the populated low-Earth-orbit band and tapering to 0.25 above approximately 2,000 km, and *A* discounts active, controllable objects to 0.40 of an equivalent defunct object.

These weights are crucial policy instruments. They would be determined and periodically reviewed by a governing body of technical and legal experts to reflect the relative importance of each risk

⁸⁰ Mark Matney, *An Overview of NASA's Orbital Debris Environment Model* (2010).

⁸¹ *Swinging for the Space Fence*, *supra* n. 43.

⁸² F. A. L. Pacheco et al., *Factor Weighting in DRASTIC Modeling*, 505 *Sci. Total Env't* 474 (2015), <https://doi.org/10.1016/j.scitotenv.2014.09.092>

driver and to steer industry behaviour. For instance, in the early phases of the fund, the governing body might place a high weight on the OPI (w_{OPI}), such as 0.5, to strongly penalize long duration missions without a credible disposal plan. This would create a powerful and immediate financial incentive for operators to comply with the 25 year deorbit guideline, effectively transforming a voluntary standard into a financially prudent business decision. Later, if spontaneous breakups of old rocket bodies become the dominant source of new debris, the weight on the Fragmentation Potential sub factor within the DGP (w_{DGP}) could be increased to incentivize the removal of these specific high risk objects.

The ODCF's primary regulatory power lies in this direct translation of risk into cost, creating a clear business case for sustainable practices. This system creates both positive and negative incentives that are directly linked to verifiable actions. When an operator successfully executes a post mission disposal manoeuvre and deorbits a satellite in compliance with international guidelines, that object's orbital lifetime, and thus its OPI, drops dramatically, causing its ORS to approach zero. This directly reduces the operator's total ORS portfolio and, consequently, their next annual contribution, providing a tangible financial return on investment for responsible end of life management. Conversely, operators who choose to fly high risk missions, for example, using satellites with a high fragmentation potential (e.g., no passivation), or placing them in crowded, high altitude orbits with long lifetimes, will be assigned a high ORS. They will pay proportionally more into the fund, effectively paying a premium for the risk they impose on the commons, which incentivizes risk mitigation at the earliest stages of mission design.

This model is inherently dynamic. The ORS for every object would be recalculated on a periodic basis. As a satellite's orbit naturally decays due to atmospheric drag, its altitude decreases. This automatically lowers its calculated OPI (as its remaining lifetime shortens) and may also lower its CPF if it moves into a less dense orbital region. This dynamic linkage ensures the risk score accurately reflects the object's current threat level, and the operator's financial contribution would automatically decrease over time in line with the object's diminishing risk.² The creation of this composite score is the central innovation, translating complex physical characteristics into a single, fungible unit of liability that allows for direct, *apples to apples* comparison of the risk posed by any two objects in orbit.²

IV. Structuring the Orbital Debris Compensation Fund (ODCF)

To design a robust and effective Orbital Debris Compensation Fund (ODCF), it is instructive to analyze the structure of successful international compensation funds established for other high risk, transnational industries, such as the International Oil Pollution Compensation (IOPC) Funds and the nuclear liability conventions.⁸³ These regimes provide proven models for collective liability, tiered compensation, and international governance that can be adapted to the unique challenges of the orbital environment.

i. The Contribution Formula

Drawing from the precedents of analogous regimes, the ODCF's contribution mechanism must be directly linked to each operator's share of the total systemic risk. While the IOPC model uses a simple proxy (tonnes of oil received),⁸⁴ and the nuclear CSC model uses a hybrid formula (installed nuclear capacity plus UN assessment rate),⁸⁵ the space debris problem allows for a more direct, physics based measure of risk via the Orbital Risk Score (ORS). The annual contribution for a given operator would be calculated as their fraction of the total aggregated risk introduced by that year's launches, multiplied by the fund's annual budgetary target. Levying contributions on the flow of newly launched objects, rather than on the entire standing population, attaches the cost to the launch decision and avoids diluting each contribution across the vast body of untraceable debris for which no operator can be charged.

Let C_i be the annual contribution for operator i , ORS_{ij} be the Orbital Risk Score for object j belonging to operator i , N_i be the total number of objects for operator i , O be the set of all operators, and F_T be the Total Annual Fund Target. The operator's annual contribution is given by the formula,

$$C_i = \left(\frac{\sum_{j=1}^{N_i} ORS_{ij}}{\sum_{k \in O} \sum_{l=1}^{N_k} ORS_{kl}} \right) \times F_T$$

⁸³ *IOPC FUNDS: Oil Reporting and Contributions*, IOPC Funds, <https://iopcfunds.org/oil-reporting-and-contributions>.

⁸⁴ *ibid.*

⁸⁵ *Nuclear Liability: A Key Component of the Public Policy Decision to Deploy Nuclear Energy in Southeast Asia*, Am. Acad. Arts & Sci., <https://www.amacad.org/publication/nuclear-liability>.

The numerator, $\sum ORS_{ij}$, represents the total risk portfolio of operator I, by adding up the ORS of all satellites that belong to operator i . The denominator, $\sum ORS_{kl}$, represents the total aggregated risk of all objects launched within the assessment window. The *Total Annual Fund Target* (F_T) would be set annually by the fund's governing body to cover administrative expenses, a reserve for anticipated compensation claims, and, critically, a budget for proactive debris removal missions.⁸⁶ This formula directly internalizes the externality of orbital risk; an operator's financial liability is precisely proportional to the risk their portfolio of assets imposes on the shared orbital environment.

ii. Fund Administration and Governance

The ODCF would require a dedicated international body for its administration, established by a new international treaty, potentially as a specialized body linked to the UN Committee on the Peaceful Uses of Outer Space (COPUOS). This governance structure would be modelled on the secretariats that administer the IOPC Funds and the role of the IAEA in the nuclear liability conventions.⁸⁷ The fund's governing body would be an Assembly of Member States, meeting periodically to make key policy decisions, such as approving the annual budget (F_T) and setting the policy driven weights for the ORS formula. A permanent, professional secretariat, composed of technical, legal, and financial experts, would be responsible for the day to day operations of the fund. The secretariat's core responsibilities would include maintaining the integrated SSA database that underpins the risk model, overseeing the transparent calculation and annual publication of ORS scores for all registered objects, administering the collection of contributions from operators and launching states, managing the claims process for damages caused by untraceable debris, and, crucially, procuring and managing contracts for Active Debris Removal (ADR) missions based on the ORS prioritization.

iii. Claims and Compensation: A No Fault Mechanism

The fund's primary compensatory function is to address the central failure of the Liability Convention: the impossibility of proving causation for damage from untraceable debris. The ODCF would operate on a no fault basis for such incidents. An operator suffering damage would file a claim with the fund. A claims commission, similar to that envisioned in the Liability Convention but with binding adjudicatory authority and funded by the ODCF, would assess the

⁸⁶ *IOPC FUNDS*, *supra* n. 73.

⁸⁷ *Nuclear Liability*, Nuclear Energy Agency (OECD), https://www.oecd-neo.org/jcms/pl_31319/nuclear-liability.

claim based on evidence that the damage was caused by an artificial, non trackable object. This decouples compensation from the burden of identifying a specific liable party, ensuring that victims of the polluted commons are compensated promptly and fairly.

Legal Feasibility, Burden-Sharing, and Incentive Effects

A proposal of this kind must answer three objections that bear directly on its legal feasibility, and not merely on its technical design. The first concerns the distribution of the burden. Because contributions are pegged to each operator's share of aggregate orbital risk, and because a small number of operators and launching States account for the overwhelming majority of the mass and the objects in orbit, the financial weight of the fund falls disproportionately on the major spacefaring States. That is, in one sense, the entire point—a polluter-pays mechanism is meant to place cost where risk is created, and the allocation tracks the responsibility already assigned to launching States by Articles VI and VII of the Outer Space Treaty. But it is also a genuine obstacle to political acceptability, and the design must accommodate it through phased entry, negotiated contribution ceilings, and the tiered burden-sharing that the oil-pollution and nuclear regimes already employ.⁸⁸

The second objection is moral hazard: that the very existence of a compensation fund might dull operators' incentives to avoid creating the small, untrackable debris for which they would otherwise escape individual liability. The objection has force, but it is mitigated by the structure of the ORS. Unlike a flat insurance premium or a fund financed by uniform per-launch levies, the ORS contribution is risk-rated *ex ante*: it charges for the creation of debris-generating risk before any collision occurs, rather than merely compensating after the fact. An operator that passivates its upper stages, deorbits promptly, and designs for low fragmentation pays measurably less, while one that does not pays more, so the marginal incentive to mitigate is preserved and indeed sharpened. The rebate and contribution-credit mechanisms described below reinforce that effect, and residual moral hazard can be addressed through experience-rating and surcharges triggered by fragmentation events.⁸⁹

⁸⁸ Because contributions track aggregate risk, and a small number of operators and launching States account for the great majority of mass and objects in orbit, the burden falls disproportionately on the major spacefaring States. This is a feature of any polluter-pays mechanism and mirrors the allocation of responsibility already implicit in Articles VI and VII of the Outer Space Treaty; it is nonetheless a genuine obstacle to political acceptability. See Andrew M. Bradley & Lawrence M. Wein, *Space Debris: Assessing Risk and Responsibility*.

⁸⁹ The objection parallels the familiar critique of liability insurance. Because ORS contributions are risk-rated *ex ante* rather than triggered only *ex post*, the fund prices the creation of debris-generating risk before any collision occurs, and experience-rating and active-debris-removal rebates preserve marginal incentives to passivate, deorbit,

The third objection is institutional. A fund of this kind cannot be conjured from the existing treaties; it would require a new instrument, most plausibly a protocol negotiated under the auspices of COPUOS, and that instrument would have to coordinate with, rather than override, the Liability Convention—preserving the absolute-liability claim for surface damage while supplying a no-fault remedy for the in-orbit harms the Convention cannot reach. None of this is trivial, and the consent-based character of international law makes it slow. But the precedents are real: the International Oil Pollution Compensation Funds and the nuclear liability conventions show that treaty-based, industry-financed compensation pools can be built and sustained for exactly the kind of diffuse, high-consequence, transnational harm that orbital debris now represents.⁹⁰

iv. Operationalizing the Framework

The establishment of the ODCF represents a significant evolution in international space law and requires a pragmatic, phased approach to implementation. Its success hinges on its ability to create tangible financial incentives that drive sustainable behaviour among all space actors.

Given the political and technical complexities, a gradual rollout is recommended to build consensus, refine the model, and demonstrate its viability. This strategy would proceed in distinct phases, each expanding the scope of the fund's application. The first phase would focus on new large satellite constellations in LEO. This cohort is an ideal starting point as the operators are few in number, their satellites are modern and well characterized, and their business models are uniquely dependent on the long term health of the very orbits they are populating, making them a highly motivated group for a pilot program.⁹¹ Following a successful pilot, the second phase would expand the system to include all new space objects launched into any Earth orbit. This would integrate the ORS assessment into the standard pre-launch authorization and licensing process for all space faring nations, ensuring that all future activities are covered from inception. The third phase would retroactively apply the model to all existing, operational satellites. This step is crucial for addressing the risk posed by the current active population, requiring operators of older, potentially higher risk satellites to begin contributing based on their assets' calculated ORS. The final and most challenging phase would address large, high risk pieces of *legacy debris*, such as

and design conservatively. Cf. Steven Shavell, On Liability and Insurance, 13 Bell J. Econ. 120 (1982).

⁹⁰ The International Oil Pollution Compensation Funds and the nuclear liability conventions (the Paris-Brussels and Vienna regimes and the Convention on Supplementary Compensation) provide established precedents for treaty-based, industry-financed compensation pools with tiered contributions and international governance.

⁹¹ Abbas, *supra* n. 3.

derelict upper stages and defunct satellites from past decades. For these objects, ORS scores would be calculated and the contribution liability would be assigned to the original launching State, creating a direct financial impetus for governments to finally address their historical contributions to the debris problem, either by paying into the fund or by funding ADR missions to remove their highest risk objects.

V. The Role of Active Debris Removal (ADR)

The ODCF provides a solution to the economic and legal paradoxes that currently stifle the development of an Active Debris Removal (ADR) market. Under the current regime, any entity attempting to remove debris assumes enormous cost and liability for cleaning a public good, with no clear financial benefit, creating a powerful disincentive for action.⁹² The ODCF can create a sustainable, market based ecosystem for ADR by directly addressing these barriers. A portion of the fund's annual budget, collected from all operators based on their risk contributions, would be allocated specifically to procure commercial ADR services. The fund's governing body, using the comprehensive ORS database, can then objectively prioritize removal targets based on which objects pose the greatest systemic risk to the orbital commons.

By ranking all derelict objects by their ORS, the fund can create a *most wanted list* and issue competitive tenders for the removal of the highest scoring objects first, thereby maximizing the risk reduction benefit for the cost incurred. This process creates a clear, predictable, and funded demand for ADR services, fostering a commercial market where companies can develop and operate technologies like robotic arms, nets, and harpoons with a viable business model.⁹³

Furthermore, the ORS model can be dynamically integrated with emerging ADR capabilities to create additional private sector incentives. For example, the fund could offer a system of rebates or future contribution credits to an operator who voluntarily contracts an ADR service to remove one of its own high ORS legacy objects. This would present the operator with a clear economic choice, continue paying a high annual contribution indefinitely for the risk posed by their derelict satellite, or make a one time capital expenditure for its removal to eliminate that future liability stream. This mechanism not only encourages the cleanup of specific high risk objects but also stimulates a secondary, private market for ADR services, complementing the publicly funded

⁹² C. Priyant Mark & Surekha Kamath, *Review of Active Space Debris Removal Methods*, 47 Space Pol'y 194 (2019).

⁹³ *ibid.*

missions initiated by the ODCF. As ADR technologies mature, this dual market approach, enabled by the financial incentives of the ORS framework, can significantly accelerate the rate at which the most dangerous legacy debris is removed from orbit, actively stabilizing the space environment.